

Airlink PPPoE Walled Garden Implementation Blueprint

Step-by-Step Architecture & Code Roadmap for Expired Redirection and Self-Care Payment Recovery

1. Executive Summary & Objective

Currently, when an Airlink PPPoE subscriber expires, the system deletes their rows from the FreeRADIUS ``radcheck`` and ``radreply`` tables. FreeRADIUS then answers incoming connection attempts with a hard **Access-Reject**. This drops the subscriber's PPP link entirely, turning off internet lights on their router and preventing them from accessing any portal to recharge.

The Goal: Transform this into a **Soft Accept (Walled Garden)**. Expired subscribers remain connected, receive an isolated IP (``10.100.32.0/20``), and get redirected by the MikroTik firewall directly to the **Nepal Airlink Recharge Portal**. Upon online payment, the billing system updates FreeRADIUS and fires a RADIUS CoA Disconnect packet (RFC 3576). The router reconnects in under 2 seconds with the active profile and full internet access.

Key Architectural Principle:

FreeRADIUS does **not** assign individual IP numbers. Instead, FreeRADIUS replies with ``Framed-Pool := pool-expired`` (or ``Mikrotik-Address-List := EXPIRED``). MikroTik handles physical IP leasing from its local hardware pools, preserving high performance and eliminating database pool desync.

2. Required Modifications Across Airlink Subsystems

Subsystem	Files / Location	Required Modifications
1. RADIUS Attributes Generator	<code>`app/Services/PppoeRadiusService.php`</code>	Add conditional attribute generation: if subscriber status is <code>`expired`</code> or <code>`suspended`</code> , set <code>`Framed-Pool := pool-expired`</code> and throttle <code>`Mikrotik-Rate-Limit := 1M/1M`</code> . If <code>`active`</code> , set <code>`Framed-Pool := pool-active`</code> and full plan bandwidth.
2. Expiration Sync Cron	<code>`app/Console/Commands/SyncPppoeStatus.php`</code>	Stop deleting <code>`radcheck`</code> / <code>`radreply`</code> rows on expiration! Instead, keep <code>`Cleartext-Password`</code> in <code>`radcheck`</code> and update <code>`radreply`</code> to the expired pool/rate limit. Fire CoA Disconnect so MikroTik immediately reconnects the user into the expired pool.
3. Recharge & Renewal Service	<code>`app/Services/PppoeRechargeService.php`</code>	When recharge succeeds (cash or online payment), rebuild <code>`radreply`</code> with <code>`pool-active`</code> and original plan bandwidth, then trigger <code>`CoaService::disconnectUsername()`</code> to restore full internet immediately.
4. MikroTik Router (NAS)	MikroTik RouterOS (<code>`/ip pool`</code> , <code>`/ip firewall`</code>)	Configure two IP pools (<code>`pool-active`</code> , <code>`pool-expired`</code>). Add DST-NAT firewall rules redirecting HTTP/HTTPS from <code>`10.100.32.0/20`</code> to the portal. Whitelist DNS (port 53) and payment gateways (eSewa, Khalti, ConnectIPS).
5. Customer Self-Care Portal	Frontend / Web Route (<code>`/portal/renew`</code>)	Portal detects subscriber identity from calling IP (<code>`radacct.framedipaddress`</code>) or token, shows due amount, processes payment gateway webhook, and triggers the Airlink recharge pipeline.

3. Concrete Code Changes in Airlink Backend

A. Update `app/Services/PppoeRadiusService.php`

```
// Inside rows(PppoeCustomer $customer):
public function rows(PppoeCustomer $customer): array {
    $isExpired = in_array($customer->status, ['expired', 'suspended']);

    // radcheck: Always allow password authentication so the router connects
    $check = [
        ['username' => $customer->username, 'attribute' => 'Cleartext-Password', 'op' => ':=', 'value' => $customer->password],
        ['username' => $customer->username, 'attribute' => 'Simultaneous-Use', 'op' => ':=', 'value' => '1']
    ];

    // radreply: Assign pool and rate limits based on subscription state
    $reply = [
        ['username' => $customer->username, 'attribute' => 'Acct-Interim-Interval', 'op' => ':=', 'value' => '300'],
        ['username' => $customer->username, 'attribute' => 'Framed-Pool', 'op' => ':=', 'value' => $isExpired ? 'pool-expired' : 'pool-active'],
        ['username' => $customer->username, 'attribute' => 'Mikrotik-Rate-Limit', 'op' => ':=', 'value' => $isExpired ? '1M/1M' : ($customer->bandwidth ? $customer->plan?->bandwidth)],
    ];
    return ['check' => $check, 'reply' => $reply];
}
```

B. Update `app/Console/Commands/SyncPppoeStatus.php`

```
// Inside handle(): When a subscriber expires, update status and radreply instead of deleting
foreach ($expiredCustomers as $customer) {
    $customer->update(['status' => 'expired']);

    // Rebuild RADIUS rows with 'pool-expired' rather than deleting credentials
    $this->customerService->rebuildRadiusRows($customer);

    // Force session drop so the router reconnects into the expired pool
    try {
        $this->coa->disconnectUsername($customer->username);
    } catch (\Throwable $e) {}
}
```

4. MikroTik RouterOS Configuration (Terminal Script)

Run these commands on your MikroTik NAS router. It creates the two pools, points expired users to the portal, and permits DNS + payment gateways so the customer can successfully complete their transaction:

```
# 1. Define the Active and Expired Pools
/ip pool add name="pool-active" ranges=10.100.0.2-10.100.31.254
/ip pool add name="pool-expired" ranges=10.100.32.2-10.100.47.254

# 2. Create Address List for Portal & Payment Gateways (eSewa, Khalti, ConnectIPS)
/ip firewall address-list add list=WALLED_GARDEN_WHITELIST address=portal.airlink.com.np
/ip firewall address-list add list=WALLED_GARDEN_WHITELIST address=esewa.com.np
/ip firewall address-list add list=WALLED_GARDEN_WHITELIST address=khalti.com

# 3. Firewall NAT: Redirect Expired HTTP traffic to Nepal Airlink Portal
/ip firewall nat add chain=dstnat src-address=10.100.32.0/20 dst-address-list=!WALLED_GARDEN_WHITELIST \
    protocol=tcp dst-port=80 action=dst-nat to-addresses=103.x.x.portal to-ports=80 comment="Airlink Portal
Redirect"

# 4. Firewall Filter: Allow DNS and Whitelisted Payment Sites, Drop General Traffic
/ip firewall filter add chain=forward src-address=10.100.32.0/20 protocol=udp dst-port=53 action=accept comment="Allow DNS"
/ip firewall filter add chain=forward src-address=10.100.32.0/20 dst-address-list=WALLED_GARDEN_WHITELIST action=accept comment="Allow Portal & Payment"
/ip firewall filter add chain=forward src-address=10.100.32.0/20 action=drop comment="Drop All Other Expired
Traffic"
```

5. Step-by-Step Rollout Checklist

Step	Action	Verification / Test
Step 1	Add `pool-active` and `pool-expired` to MikroTik NAS along with the NAT redirect and gateway whitelist rules.	Verify on MikroTik with `/ip pool print` and `/ip firewall nat print`.
Step 2	Update `PppoeRadiusService.php` to include `Framed-Pool := pool-expired` for expired accounts.	Run `radtest` against an expired test account (`KHPPOE00002`). Verify `Framed-Pool = pool-expired` in Access-Accept.
Step 3	Update `SyncPppoeStatus.php` cron so that expiration re-writes `radreply` with the expired profile and issues a CoA disconnect.	Run `php artisan pppoe:sync-status` on a test customer. Check that `radcheck` password remains intact and session is kicked.
Step 4	Test self-care recharge flow. Connect a test router to the expired pool, pay via portal, verify CoA disconnect fires, and confirm full speed restoration.	Router automatically reconnects into `10.100.0.0/19` with full internet access in under 2 seconds.